

Practical 8: ARIMA Modeling of AirPassengers Dataset

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1. Question

Consider the “AirPassengers” data from R library and write the R program for the following:

- (f) Convert the data into a time series object.
 - (g) Plot the data to identify the dominant component.
 - (h) Decompose the data to observe the dominating components more clearly.
 - (i) Check stationarity or non-stationarity using ACF/PACF plot.
 - (j) Check stationarity or non-stationarity using the KPSS test.
 - (k) If data is non-stationary, make it stationary using an appropriate operator.
 - (l) Based on the dominating component, select the suitable technique to fit the data.
 - (m) Fit the data using the selected model and estimate the parameters of the model.
 - (n) Check the goodness of fit of the model.
-

2. R Code (Question-wise Solution)

```
# To delete previous history or stored variables
rm(list = ls(all = TRUE))

library(tseries)
library(forecast)
library(datasets)

# -----
# (f) Convert the data into a time series object
```

```
# -----

data(AirPassengers)

ap_data <- as.ts(AirPassengers)

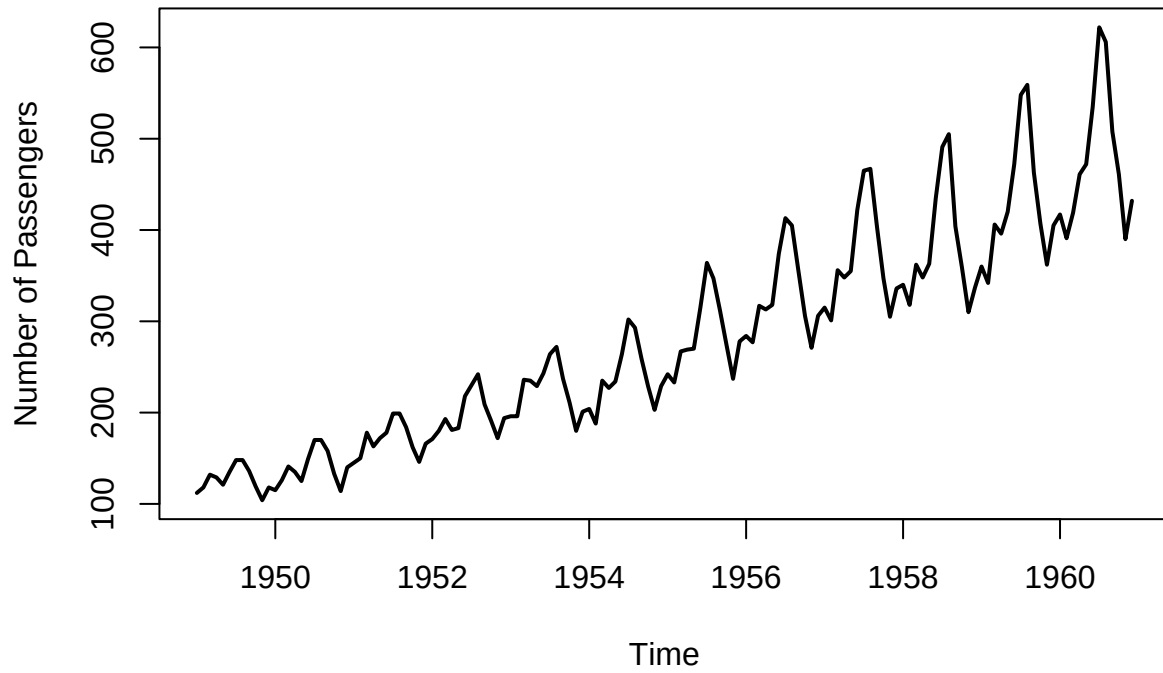
# Printing the data
ap_data
```

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
## 1949	112	118	132	129	121	135	148	148	136	119	104	118
## 1950	115	126	141	135	125	149	170	170	158	133	114	140
## 1951	145	150	178	163	172	178	199	199	184	162	146	166
## 1952	171	180	193	181	183	218	230	242	209	191	172	194
## 1953	196	196	236	235	229	243	264	272	237	211	180	201
## 1954	204	188	235	227	234	264	302	293	259	229	203	229
## 1955	242	233	267	269	270	315	364	347	312	274	237	278
## 1956	284	277	317	313	318	374	413	405	355	306	271	306
## 1957	315	301	356	348	355	422	465	467	404	347	305	336
## 1958	340	318	362	348	363	435	491	505	404	359	310	337
## 1959	360	342	406	396	420	472	548	559	463	407	362	405
## 1960	417	391	419	461	472	535	622	606	508	461	390	432

```
# -----
# (g) Plot the data to identify dominant component
# -----

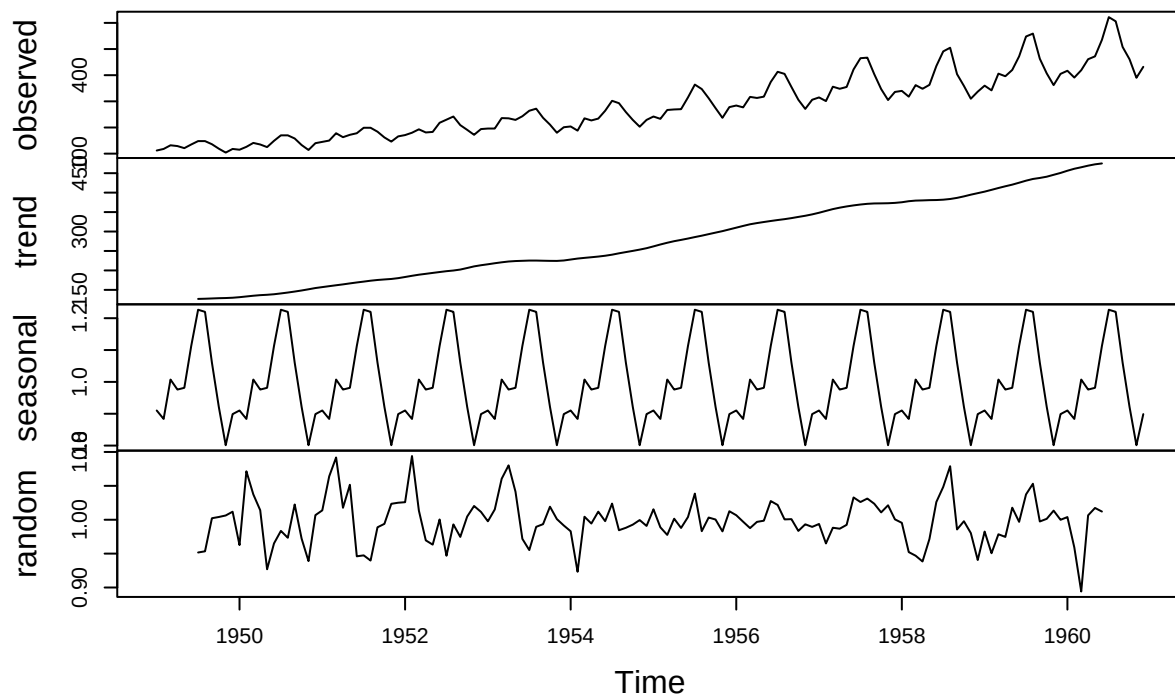
plot(ap_data,
      main = "AirPassengers Data",
      xlab = "Time",
      ylab = "Number of Passengers",
      lwd = 2)
```

AirPassengers Data



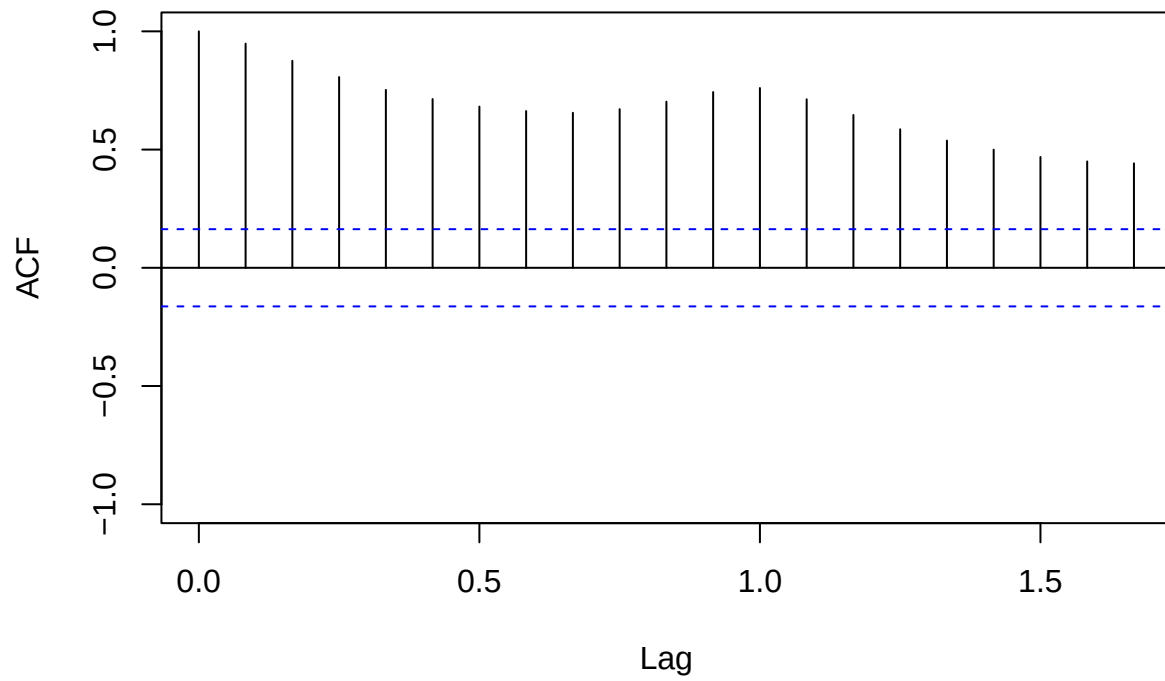
```
# -----  
# (h) Decompose the data  
# -----  
  
ap_decomp <- decompose(ap_data, type = "multiplicative")  
plot(ap_decomp)
```

Decomposition of multiplicative time series



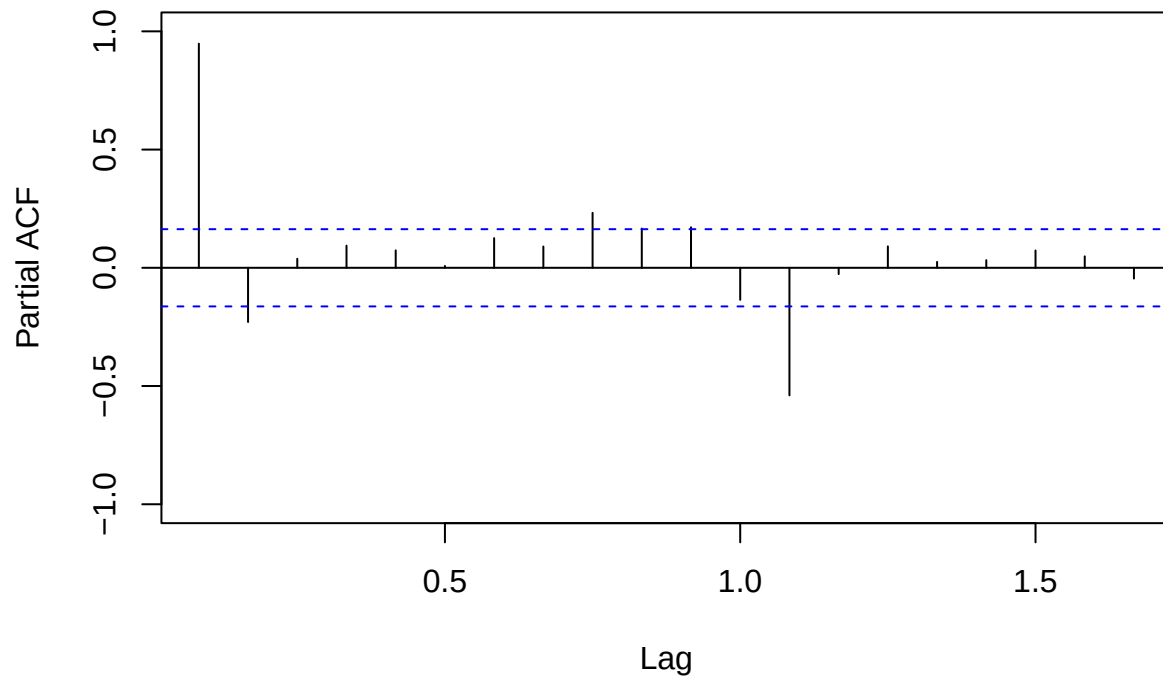
```
# -----  
# (i) Check stationarity using ACF/PACF  
# -----  
  
acf(ap_data,  
    main = "ACF of AirPassengers Data",  
    lag.max = 20,  
    ylim = c(-1,1))
```

ACF of AirPassengers Data



```
pacf(ap_data,  
      main = "PACF of AirPassengers Data",  
      lag.max = 20,  
      ylim = c(-1,1))
```

PACF of AirPassengers Data



```
# -----  
# (j) Check stationarity using KPSS Test  
# -----  
  
kpss.test(ap_data)
```

```
## Warning in kpss.test(ap_data): p-value smaller than printed p-value
```

```
##  
## KPSS Test for Level Stationarity  
##  
## data: ap_data  
## KPSS Level = 2.7395, Truncation lag parameter = 4, p-value = 0.01
```

```
# -----  
# (k) If non-stationary, make it stationary  
# Use log transformation (variance stabilization)  
# followed by differencing  
# -----  
  
ap_log <- log(ap_data)  
  
ap_d1 <- diff(ap_log)  
  
kpss.test(ap_d1)
```

```
## Warning in kpss.test(ap_d1): p-value greater than printed p-value
```

```
##
```

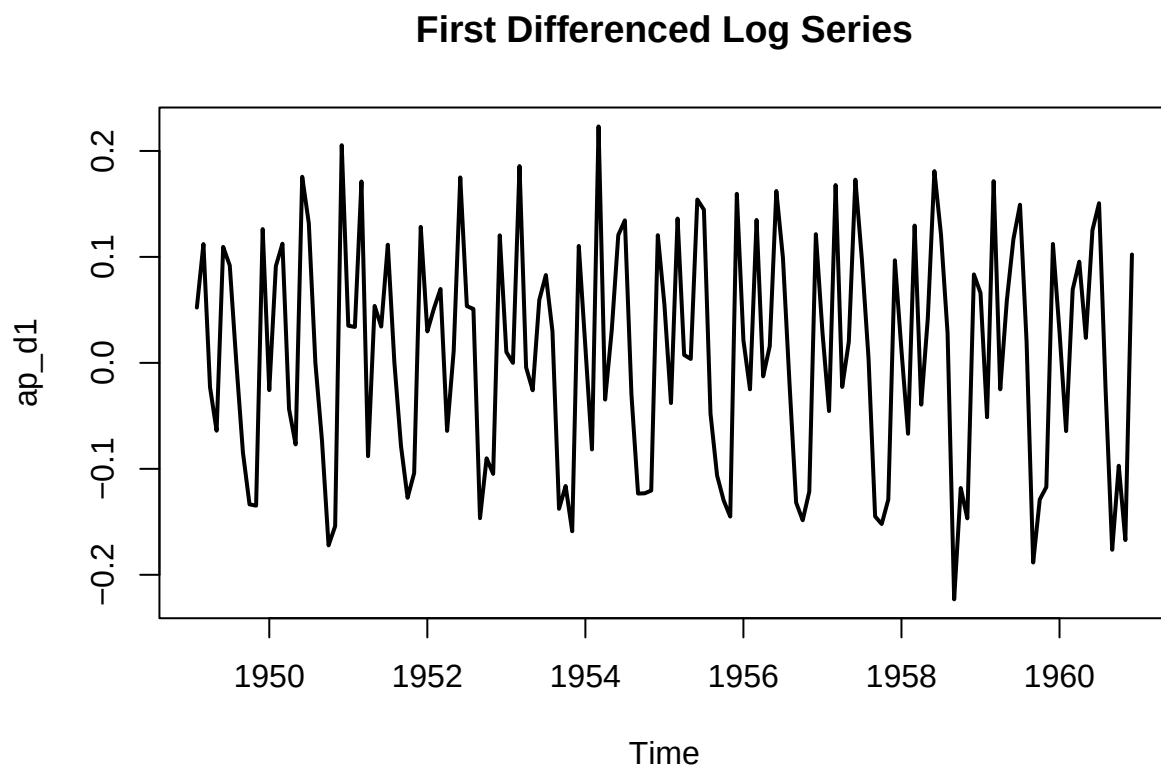
```
## KPSS Test for Level Stationarity
```

```
##
```

```
## data: ap_d1
```

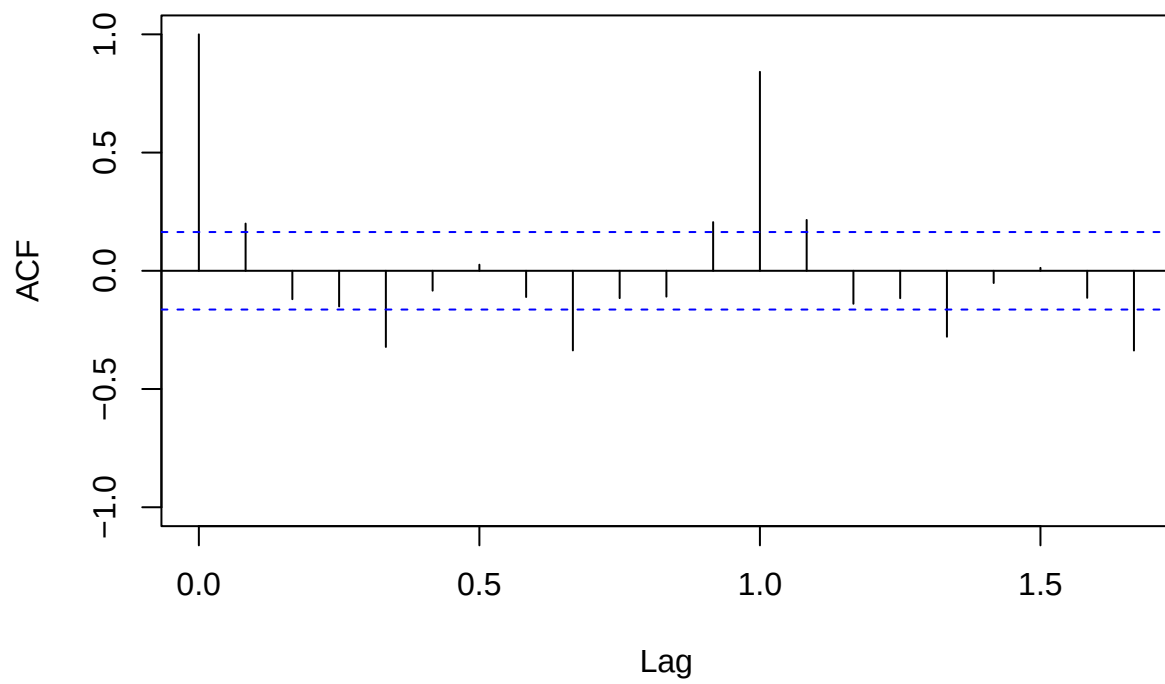
```
## KPSS Level = 0.028205, Truncation lag parameter = 4, p-value = 0.1
```

```
plot(ap_d1,  
     main = "First Differenced Log Series",  
     xlab = "Time",  
     lwd = 2)
```



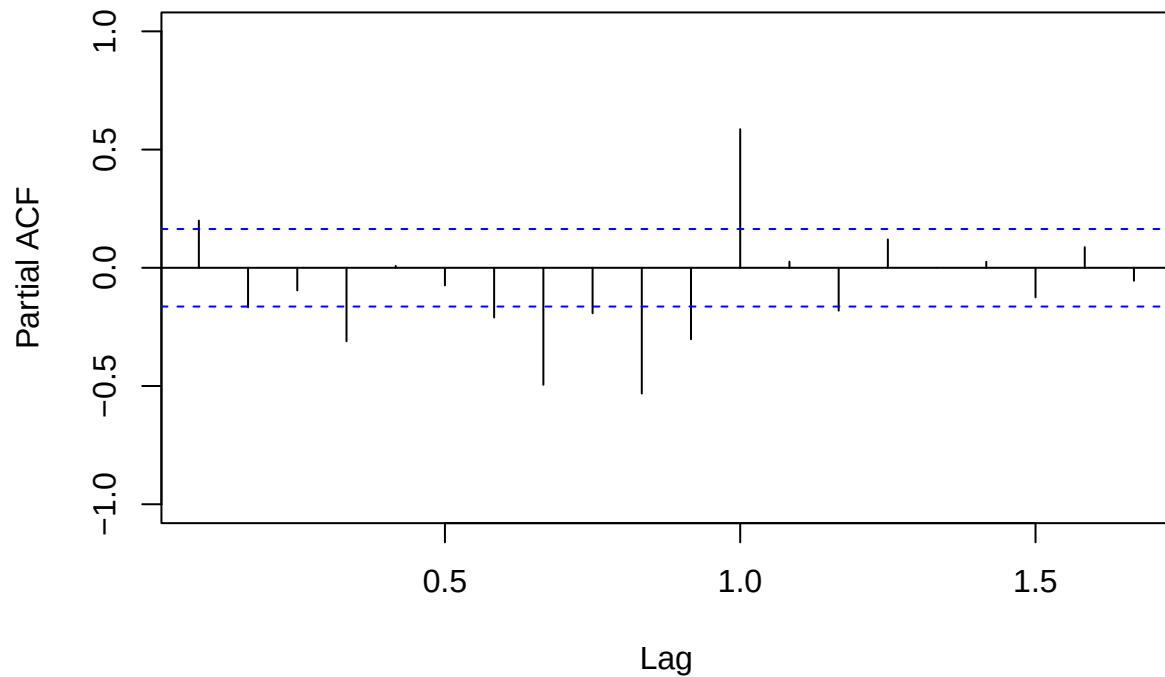
```
acf(ap_d1,  
     main = "ACF of Differenced Log Series",  
     lag.max = 20,  
     ylim = c(-1,1))
```

ACF of Differenced Log Series



```
pacf(ap_d1,  
      main = "PACF of Differenced Log Series",  
      lag.max = 20,  
      ylim = c(-1,1))
```


PACF of Differenced Log Series



```
# -----
# (l) Model Identification using ACF/PACF and AIC
# Since trend and seasonality are dominant,
# we use ARIMA model
# -----
```

```
ap_fit <- auto.arima(ap_log)
```

```
ap_fit
```

```
## Series: ap_log
## ARIMA(0,1,1)(0,1,1)[12]
##
## Coefficients:
##          ma1      sma1
##       -0.4018  -0.5569
## s.e.   0.0896   0.0731
##
## sigma^2 = 0.001371: log likelihood = 244.7
## AIC=-483.4   AICc=-483.21   BIC=-474.77
```

```
# -----
# (m) Fit the selected model and estimate parameters
# (Model fitted above using auto.arima)
# -----
```

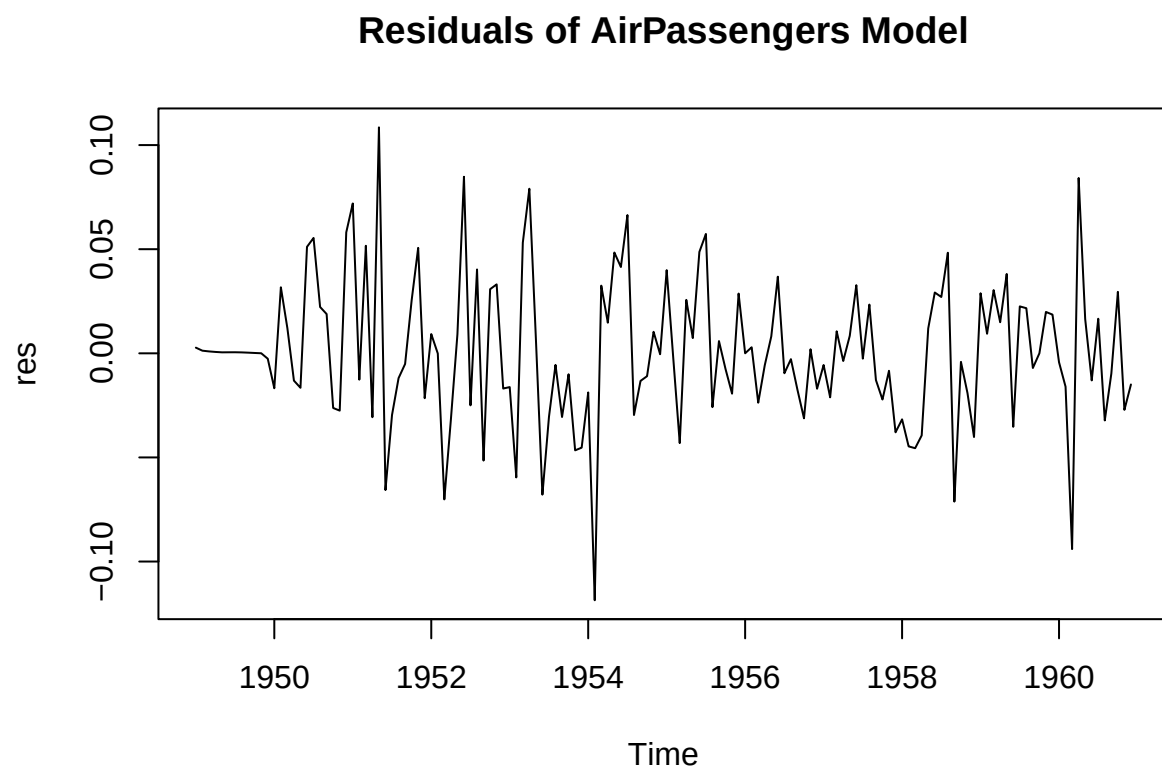
```

# -----
# (n) Check Goodness of Fit (Diagnostic Checking)
# -----

res <- ap_fit$residuals

# Plot residuals
plot(res,
      main = "Residuals of AirPassengers Model")

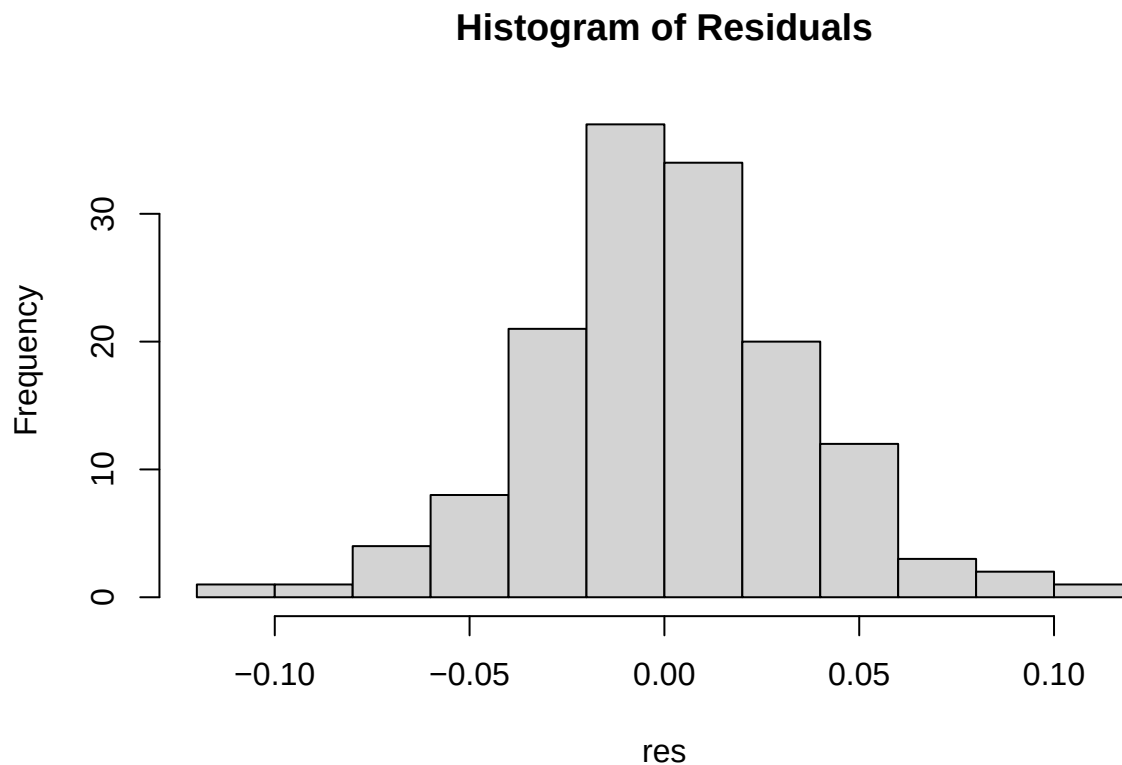
```



```

# Check Normality
hist(res,
      main = "Histogram of Residuals")

```



```
shapiro.test(res)
```

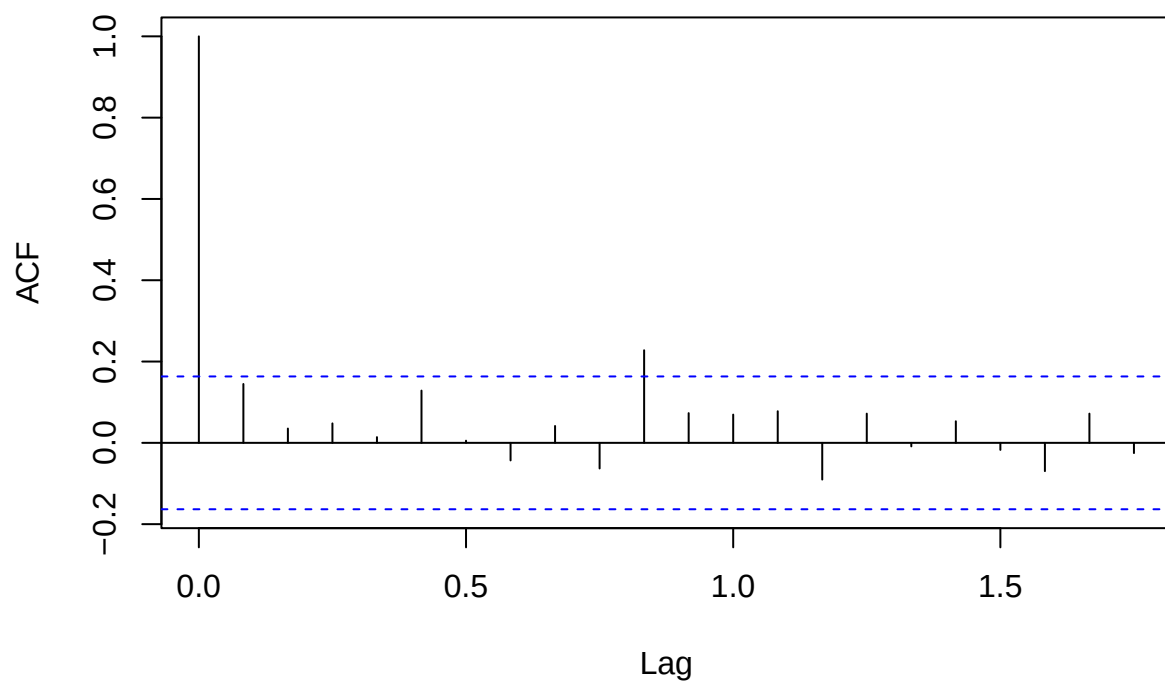
```
##  
## Shapiro-Wilk normality test  
##  
## data:  res  
## W = 0.98637, p-value = 0.1674
```

```
# Check Serial Correlation  
Box.test(res,  
  lag = 20,  
  type = "Box-Pierce")
```

```
##  
## Box-Pierce test  
##  
## data:  res  
## X-squared = 16.062, df = 20, p-value = 0.7128
```

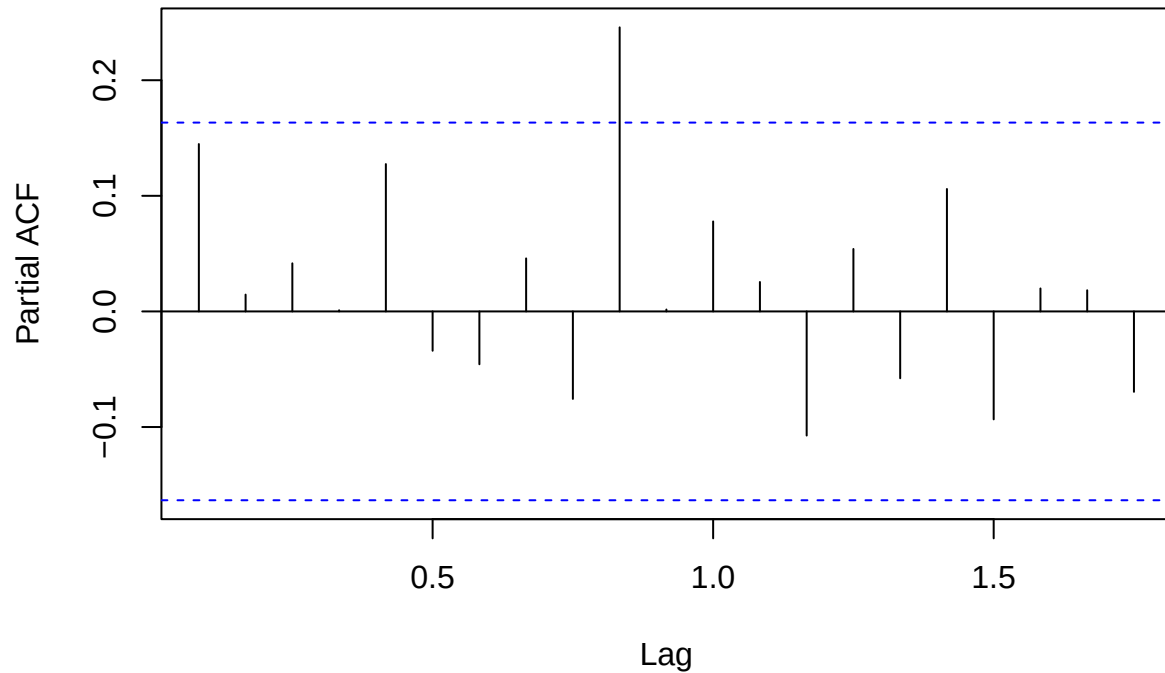
```
# Check Heteroscedasticity  
acf(res^2,  
  main = "ACF of Squared Residuals")
```

ACF of Squared Residuals



```
pacf(res^2,  
      main = "PACF of Squared Residuals")
```

PACF of Squared Residuals



Conclusion

The AirPassengers data shows a strong upward trend and multiplicative seasonality. ACF and KPSS test confirmed that the original series is non-stationary. After applying log transformation and first differencing, the series became stationary. Based on trend and seasonal dominance, an ARIMA model was selected using AIC. Diagnostic checking confirmed:

- Residuals are approximately normal
- No serial correlation
- Constant variance

Hence, the selected ARIMA model is a good fit for the AirPassengers data.